



National University of Sciences & Technology (NUST)
School of Electrical Engineering and Computer Science (SEECs)
Department of Computing

CS-837 Advanced Operating Systems

Course Code:	CS-837	Semester:	1st
Credit Hours:	3+0	Prerequisite Codes:	N/A
Instructor:	Dr. Sarosh Tahir	Class:	MS CS / IT / DS / AI
Office:	ONT/SDON Lab	Telephone:	N/A
Lecture Days:	3 × 50-minute lectures/week	E-mail:	sarosh.tahir@seecs.edu.pk
Class Room:	CR-3	Consulting Hours:	Flexible / by appointment
Lab Engineer:	N/A	Lab Engineer Email:	N/A
Knowledge Group:	Core Computer Science	Updates on LMS:	After every lecture

1. Course Description

CS-837 is an advanced study of operating-system design, mechanisms, policies, and contemporary systems practice. The course revisits classical abstractions—processes, threads, synchronization, scheduling, memory, storage, I/O, protection, and distributed processing—at a deeper level and connects them to multicore/NUMA machines, virtualization, containers, cloud and datacenter resource management, programmable kernels, heterogeneous accelerators, and modern security/isolation mechanisms. Emphasis is placed on architectural trade-offs, performance, scalability, correctness, isolation, and the interaction between hardware and the operating-system kernel.

2. Course Objectives

- Develop a rigorous understanding of advanced operating-system abstractions and kernel mechanisms.
- Analyze performance, scalability, fairness, isolation, and reliability trade-offs in modern operating systems.
- Relate classical OS concepts to multicore, NUMA, virtualized, containerized, cloud, and heterogeneous computing environments.
- Evaluate contemporary OS designs and research through analytical problems, case studies, technical readings, and a term project.
- Use systems-oriented tools and programming environments, where appropriate, to support theoretical analysis and experimental validation.

3. Course Learning Outcomes (CLOs)

CLO	Learning Outcome	Bloom's Level
CLO-1	Explain and analyze advanced OS abstractions, kernel structures, processes/threads, concurrency, scheduling, memory, storage, and I/O mechanisms.	C4 – Analyze
CLO-2	Evaluate synchronization, multicore scheduling, memory-management, virtualization, and isolation techniques with respect to correctness, performance, scalability, and resource efficiency.	C5 – Evaluate
CLO-3	Compare and assess modern OS approaches for virtualization, containers, distributed/cloud systems, security, and heterogeneous/accelerator-aware computing.	C5 – Evaluate
CLO-4	Design and justify a systems-oriented solution or research investigation that applies advanced OS concepts using appropriate contemporary tools and evidence.	C6 – Create

4. Assessment Scheme

Assessment	Weight	Planned Quantity / Scope
End Semester Examination (ESE)	40%	Comprehensive
Mid Semester Examination (MSE)	25%	Pre-MSE content
Quizzes	15%	6 quizzes
Assignments	10%	4 assignments
Term Project	10%	Research/system-oriented project with report and presentation/demo as appropriate



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Assessment	Weight	Planned Quantity / Scope
Total	100%	

5. Indicative CLO–Assessment Mapping

Assessment	CLO-1	CLO-2	CLO-3	CLO-4
Quizzes	✓	✓	✓	
Assignments	✓	✓	✓	✓
MSE	✓	✓		
Term Project		✓	✓	✓
ESE	✓	✓	✓	

6. Week-by-Week Delivery and Assessment Plan

Week	Dates	Teaching Plan	Assessment / Milestone
1	7–11 Sep	L1–L3: Modern OS foundations, architectures, kernel interfaces	Course orientation
2	14–18 Sep	L4–L6: Processes, threads, multicore performance/Amdahl's Law	Quiz 1
3	21–25 Sep	L7–L9: Race conditions, atomic primitives, synchronization abstractions	Assignment 1 issued
4	28 Sep–2 Oct	L10–L12: Classical problems, scalable synchronization, memory consistency	Quiz 2
5	5–9 Oct	L13–L15: Scheduling foundations, multiprocessor scheduling, load balancing	Assignment 1 due
6	12–16 Oct	L16–L18: Real-time, NUMA/heterogeneous scheduling, VM architecture	Quiz 3; Assignment 2 issued
7	19–23 Oct	L19–L21: TLBs, paging, allocation/kernel memory	
8	26–30 Oct	L22–L25: Replacement, NUMA memory, deadlocks, advanced resource control	Assignment 2 due; MSE review
9	2–8 Nov	MID SEMESTER EXAMINATION (MSE)	25%
10	9–13 Nov	L26–L28: Virtualization fundamentals, CPU and memory virtualization	Quiz 4
11	16–20 Nov	L29–L31: I/O virtualization, migration/resource management, containers	Assignment 3 issued
12	23–27 Nov	L32–L34: Container internals/security context, modern file-system architecture	Term Project proposal/topic approval
13	30 Nov–4 Dec	L35–L37: Crash consistency, modern storage, high-performance I/O	Quiz 5; Assignment 3 due
14	7–11 Dec	L38–L40: Distributed OS, RPC/time/synchronization, consistency/replication	Assignment 4 issued
15	14–18 Dec	L41–L43: Protection/isolation, modern security, cloud/datacenter OS	Quiz 6; Project checkpoint
16	21–25 Dec	L44: Accelerators/eBPF/observability + project consultation/reading discussion	Holiday-sensitive week
17	28 Dec–3 Jan	L45: Emerging OS/AI/disaggregated computing; project presentations and ESE synthesis	Assignment 4 due; Term Project due/presentation
18	4–10 Jan	END SEMESTER EXAMINATION (ESE)	40%



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7. Textbooks and Reading List

Type	Reference	Use
Primary	William Stallings, Operating Systems: Internals and Design Principles.	Core architecture, processes, concurrency, scheduling, memory, I/O, file systems, security and distributed processing.
Primary / Open	Remzi H. Arpaci-Dusseau and Andrea C. Arpaci-Dusseau, Operating Systems: Three Easy Pieces (OSTEP).	Processes, virtualization, concurrency, persistence; conceptual and systems-oriented reading.
Reference	Andrew S. Tanenbaum and Herbert Bos, Modern Operating Systems.	Alternative treatment of processes, memory, file systems, virtualization, distributed and security topics.

8. Course Policies

Quiz Policy

Six quizzes are planned. Quizzes may test material from recent lectures, assigned readings, and analytical examples. The instructor may announce the exact format and timing through the LMS.

Assignment Policy

Four assignments are planned. Submission instructions, due dates, collaboration rules, and any required source/code artifacts will be communicated through the LMS.

Term Project Policy

Project topics require instructor approval. Students must clearly distinguish their own contribution from reused code, datasets, tools, or prior work and must cite all external sources.

Academic Integrity

Academic integrity requirements of NUST/SEECS apply. Plagiarism, unauthorized collaboration, fabrication of results, or misrepresentation of contributions may result in penalties under applicable university policy.

Attendance

Attendance requirements will follow current NUST/SEECS academic regulations and policies.